

Barath T

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EDUCATION

Vellore Institute of Technology (VIT) – Chennai

Bachelor of Technology in Computer Science and Engineering (Data Science)

2024 – 2028 (Expected)

CGPA: 7.99

EXPERIENCE

Artificial Intelligence and Machine Learning Intern

BlackVolt Technologies Pvt. Ltd.

June 2026 – Sept 2026

Chennai, India

- Researched critical AUV parameters and conducted a comparative study of DAVE and Stonefish, focusing on hydrodynamic modelling, simulation architecture, and relevant inputs/outputs.
- Studied Reinforcement Learning fundamentals including MDPs, Q-learning, policies, and exploration–exploitation using Aurélien Géron’s Hands-On Machine Learning, with practical implementation through chapter exercises.

Technical Department Member

Zero Bugs Club, VIT Chennai

Sep 2026 – Present

Chennai, India

- Contributed to the development and maintenance of full-stack web applications, working across frontend interfaces, backend services, APIs, and database integrations.
- Collaborated on software development using Git/GitHub workflows, including issue tracking, feature branches, pull requests, code reviews, and CI/CD validation.
- Implemented and refined features, APIs, and application functionality based on project requirements while following collaborative development and code quality practices.

PROJECTS

Unified Project Management System | React, Node.js, Express, MongoDB

- Engineered a full-stack capstone platform for project tracking, guide allocation, and review workflows; implemented JWT authentication and role-based access control.
- Designed a randomized panel-assignment algorithm preventing duplicate guide/panel allocation, ensuring fair, conflict-free evaluations.
- Co-inventor, Published Indian Patent Application for the system, filed under faculty supervision.

MemoryMap: Organizational Knowledge platform | React, FastAPI, RAG, Neo4j, ChromaDB

[GitHub](#)

- Built an AI-powered organizational knowledge platform using hybrid RAG, combining semantic vector search and Neo4j knowledge graphs to connect fragmented organizational information and provide context-aware answers.
- Developed modular FastAPI backend pipelines for document ingestion, chunking, embedding generation, AI-based entity/relationship extraction, vector retrieval, and graph-based retrieval.
- Integrated GitHub as a knowledge source and implemented ChromaDB + Neo4j storage pipelines to link unstructured documents with structured entities, relationships, and contextual evidence.

Power BI Analytics Dashboard Suite | PowerBI, Power Query, DAX, Excel, Python

[Github](#)

- Developed interactive Power BI dashboards for retail sales, customer churn, and marketing analytics.
- Performed data cleaning, transformation, and KPI development to generate actionable business insights.
- Built dynamic visualizations and reports to support data-driven decision-making across multiple business domains.

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, Java, C, C++, JavaScript

Backend: FastAPI, Django, Node.js, Express, REST API

Frontend: React, HTML, CSS, Vite, Framer Motion

Database: PostgreSQL, MongoDB, MySQL, ChromaDB, Oracle, Neo4j

AI / ML: RAG Pipelines, SentenceTransformers, LLM Integration

Tools: Git, GitHub, Postman, Docker, Linux, VS Code, PowerBI

Certifications: HackerRank Skill Verifications, MERN Stack (FACE Prep), Data Science & Analytics (Future Interns), ‘Agentic AI: Build your first Agentic AI System’ and ‘Introduction to Prompt Engineering for Generative AI’ (LinkedIn Learning), Data Science (Launched Global)

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